



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- SpaceX reduces launch costs by reusing the first stage of its rockets. However, successful landing and reuse depend on mission conditions such as payload and orbit.
- This project applies machine learning techniques to publicly available launch data to predict whether the first stage will be successfully reused. The model can help analyze launch outcomes and demonstrate the use of predictive analytics in aerospace applications.
- We used four classification approaches and achieved 83% accuracy from all the approaches. The approaches used are: Logistic Regression, Support Vector Machine (SVM), Decision Tree method, and K-NN method. We also have performed exploratory data analysis and presented the interactive visual representation of the SpaceX launch statistics.

Introduction

- SpaceX provides satellite internet services, supports missions to the International Space Station, and launches satellites into space using reusable rockets.
- The cost of rocket launches by SpaceX is 2.6 times less than other providers because SpaceX reuses the the first stage, which is very expensive. But sometimes the first stage does not land, it may crash or the company scarifies the first stage due to the mission parameters like payload, orbit, and customer.
- The problem is to determine whether SpaceX will reuse the first stage using a machine learning model and use public information to predict if SpaceX will reuse the first stage.

Section 1

Methodology

Methodology

- Data collection
- Data wrangling
- Exploratory data analysis (EDA) using visualization and SQL
- Interactive visual analytics using Folium and Plotly Dash
- Predictive analysis using classification models

Data Collection

- SpaceX launch data was collected using SpaceX REST API. The data includes rockets used, payload delivered, launch specifications, and landing outcome. GET method requests library to uses URL that points to RESTAPI end points to download the data.
- We also used BeautifulSoup to scrap html table data form SpaceX Wikipedia page.
- The values from the table are stored in a Python dictionary and converted into a Python DataFrame.

Data Collection – SpaceX API

- GET method is called with the URL of the launch data REST API end point using Python requests library.
- Received json response is transformed into a dataframe using json_normalize method
- Jupyter notebooks developed for this project can be accessed from:
- <https://github.com/gandhiUH/Data-Science-Capstone-.git>
- <https://github.com/gandhiUH/Data-Science-Capstone-/blob/main/jupyter-labs-webscraping.ipynb>

API :

url =
"https://api.spacexdata.com/v4/launches"

response = requests.get(url)

data = response.json()

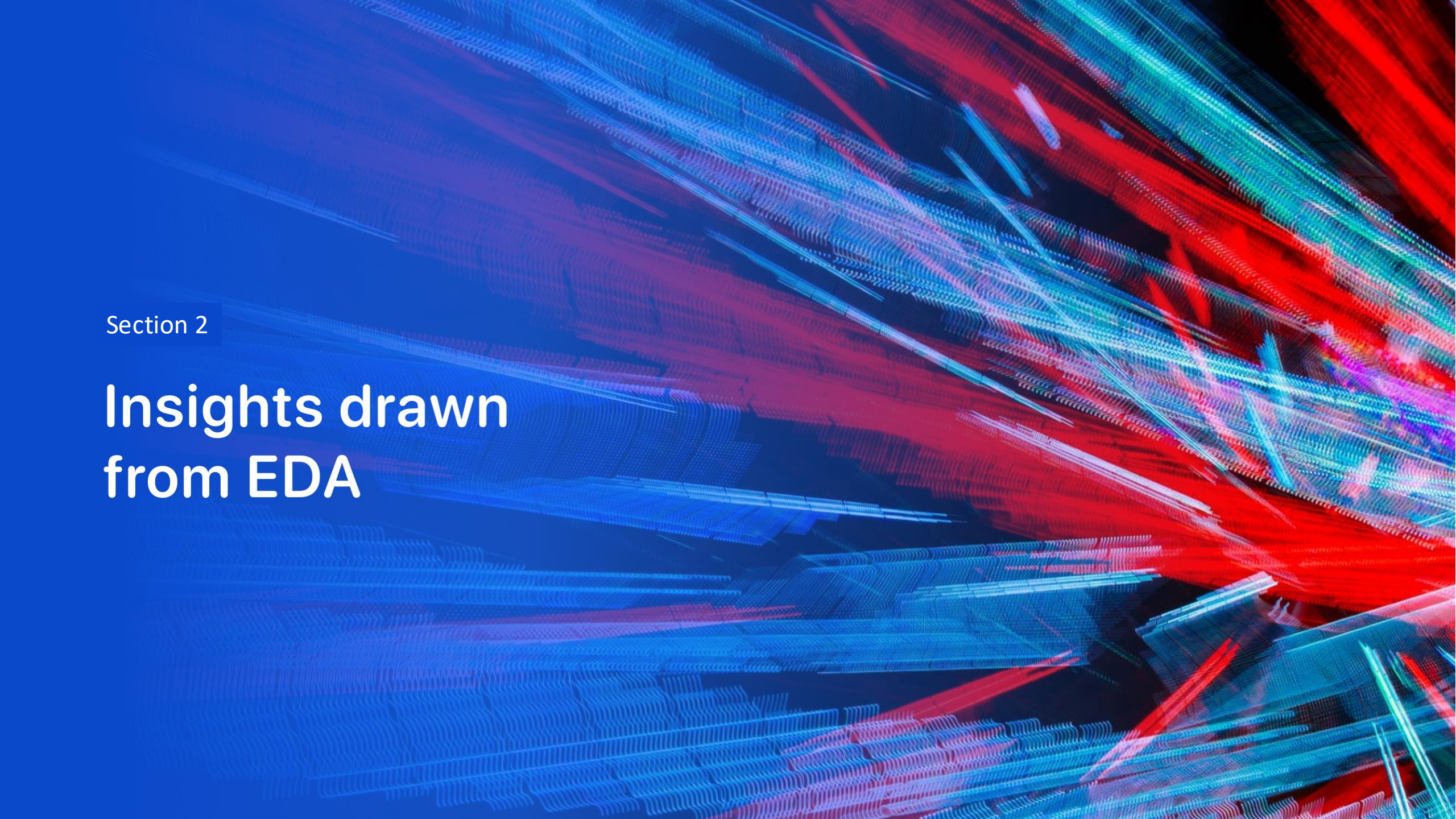
df = pd.json_normalize(data)

Data Collection – Web Scraping

1. `static_url:`https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922
2. `response = requests.get(static_url, headers=headers)`
3. `soup = BeautifulSoup(response.content, "html.parser")`
4. `html_tables = soup.find_all('table')`
5. `first_launch_table = html_tables[2]`
6. Find the column names using 'th' tag, create a dictionary with keys using column names
7. Iterate through the launch table for each row using 'td' tags, parse each column and append them to the dictionary. From the dictionary, data frame is created.
8. `df= pd.DataFrame({ key:pd.Series(value, dtype='object') for key, value in launch_dict.items() })`

Data Wrangling

- Data frame was created by reading the dataset_part_1.csv
- `df=pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv")`
- Missing values in all features are checked
- Number of launches in each site was calculated.
- Occurrences of each orbit was calculated.
- Class label feature column was with 1's and 0's created based Landing outcome's success or failure.
- Reference: <https://github.com/gandhiUH/DataScience-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. Overlaid on these streaks is a faint, light blue grid pattern, giving the impression of a digital or data-driven environment.

Section 2

Insights drawn from EDA

EDA with SQL

- **Queries used for exploratory Data Analysis**
- `%sql` `sqlite:///my_data1.db`
- `%sql` `create table SPACEXTABLE as select * from SPACEXTBL where Date is not null`
- `%sql` `SELECT DISTINCT "Launch_Site" FROM SPACEXTBL`
- `%sql` `SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%'`
- `%sql` `SELECT DISTINCT "Landing_Outcome" FROM SPACEXTABLE`
- `%sql` `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE "CUSTOMER" = "NASA (CRS)"`
- `%sql` `SELECT AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'`
- `%sql` `SELECT MIN(Date) FROM SPACEXTABLE WHERE "Landing_Outcome" = "Success (ground pad)"`

EDA With SQL

- `%sql SELECT "Booster_Version" from SPACEXTABLE WHERE "Landing_Outcome" = "Success (drone ship)" AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000`
- `%sql SELECT "Mission_Outcome", COUNT(*) From SPACEXTABLE GROUP BY "Mission_Outcome"`
- `%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" =(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)`
- `%%sql SELECT CASE substr(Date, 6,2) WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March' WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June' WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September' WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December' END AS "Month_Name", "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr(Date, 1, 4) = '2015';`

EDA With SQL

- `%sql SELECT "Landing_Outcome", COUNT(*) AS "Outcome_Count" FROM SPACEXTABLE WHERE Date Between '2010-06-04' and '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY "Outcome_Count" DESC`

Launch Site Names Begin with 'CCA'

- Launch site CCAFS LC 40 used only mainly LEO orbit, one HEO orbit, and a few time GTO orbit with HEO and LEO with small satellites (less than 3200 kg of payload), and payloads between 3000 -4700 kg at GTO orbit.
- Site CCAFS SLC 40 has more landing success rates than CCAFS LC 40

```
26]: %sql SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

```
* sqlite:///my_data1.db  
Done.
```

```
26]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG__	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

All Launch Site Names

- There are 4 launch sites given in the dataset.

```
In [11]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTBL
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[11]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is 2928.4 kg

```
[33]: %sql SELECT AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
      * sqlite:///my_data1.db
      Done.
[33]: AVG(PAYLOAD_MASS__KG_)
      2928.4
```


Total Payload Mass

- Total payload carried by NASA missions is 45596 kg.

```
[15]: %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE "CUSTOMER" = "NASA (CRS)"
      * sqlite:///my_data1.db
      Done.
[15]: SUM(PAYLOAD_MASS__KG_)
      45596
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 is shown in the screen shot.

```
[50]: %sql SELECT "Booster_Version" from SPACEXTABLE WHERE "Landing_Outcome" = "Success (drone ship)" AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
* sqlite:///my_data1.db
Done.
[50]: Booster_Version
```

F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

First Successful Ground Landing Date

- The first successful landing outcome on ground pad was on 12-22-2015.

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[48]: %sql SELECT MIN(Date) FROM SPACEXTABLE WHERE "Landing_Outcome" = "Success (ground pad)"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[48]: MIN(Date)
```

```
2015-12-22
```

Boosters Carried Maximum Payload

- The names of the booster which have carried the maximum payload mass of 15600 Kg was is shown in the screenshot below
- It is interesting know all of them(12) were in LEO orbit and except two of them, remaining 10 are landed successfully

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" =(SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE)
* sqlite:///my_data1.db
Done.
: Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7
```

2015 Launch Records

- Failed landing outcomes in drone ship in year 2015
- There are two failures with the booster version of F9v1.1 B1012 at launch site CCAFS LC-40

```
[21]: %%sql
SELECT CASE substr(Date, 6,2) WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March'
      WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June'
      WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September'
      WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December'
      END AS "Month_Name",
      "Landing_Outcome",
      "Booster_Version",
      "Launch_Site"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Failure (drone ship)'
AND substr(Date, 1, 4) = '2015';
```

```
* sqlite:///my_data1.db
Done.
```

```
[21]:
```

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:
- Success and failure attempts are the same 5 with droneship

```
[22]: %sql SELECT "Landing_Outcome", COUNT(*) AS "Outcome_Count" FROM SPACEXTABLE WHERE Date Between '2010-06-04' and '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY "Outcome_Count" DESC
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[22]:
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes:
 - Success = 99
 - Failures = 1

```
[54]: %sql SELECT "Mission_Outcome", COUNT(*) From SPACEXTABLE GROUP BY "Mission_Outcome"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[54]:
```

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Results From SQL EDA

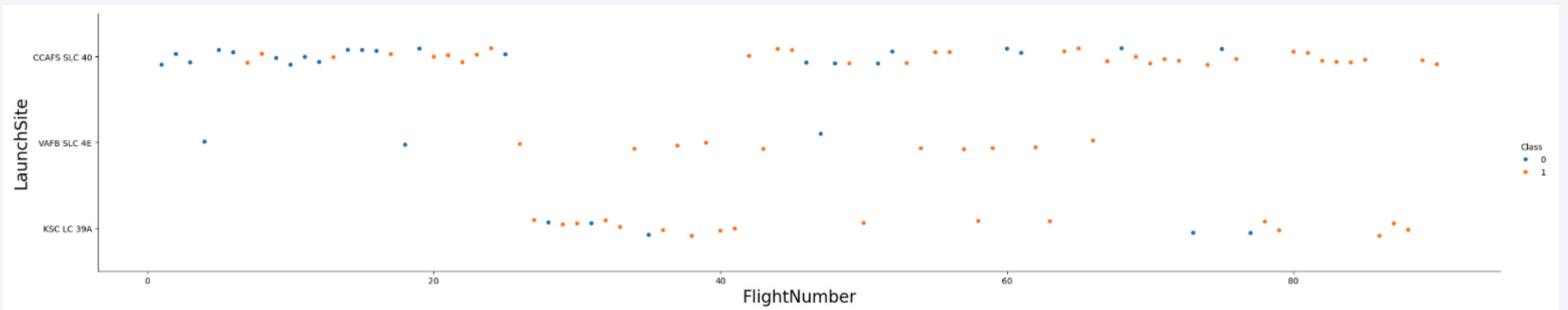
- From the SQL analysis the following are the notable insights:
- There are 4 launch sites given in the dataset
- Total payload carried by NASA missions is 45596 kg.
- The average payload mass was 2928.4 kg
- Total number of successful and failure mission outcomes:
 - Success = 99
 - Failures = 1

EDA with Data Visualization

- We used Matplotlib, Folium Map, Dash, and Plotly libraries for EDA with Visualization.
- Following static charts are created using Matplotlib:
 - Flight Number Vs Pay load mass
 - Flight Number Vs Launch site
 - Pay load mass Vs Launch site
 - Orbit Vs Success rate
 - Orbit Vs Flight number
 - Orbit Vs. Pay load mass
 - Success trend over the years.

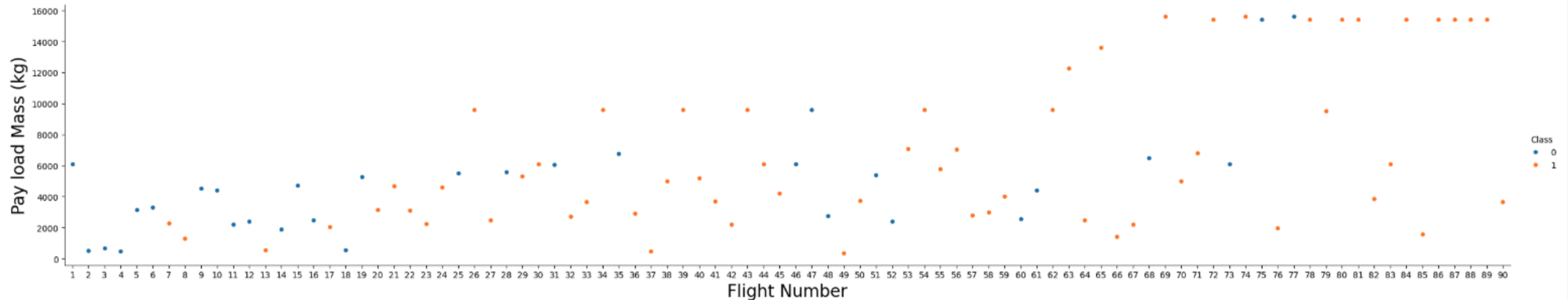
Flight Number vs. Launch Site

- CCAFS SLC 40 Launch site has more flights sent. The success rate increased as number of flights sent increased.
- VAFB SLC4E site has sent minimum number of flights



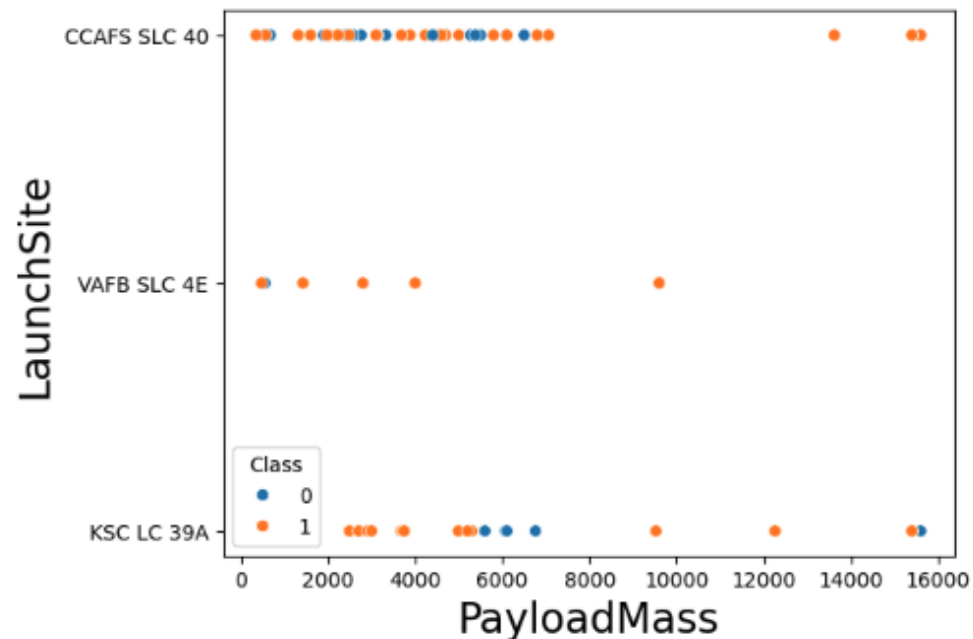
Payload vs. Launch Site

- As the flight number increases, the first stage is more likely to land successfully.
- The payload mass also appears to be a factor; even with more massive payloads, the first stage often returns successfully.



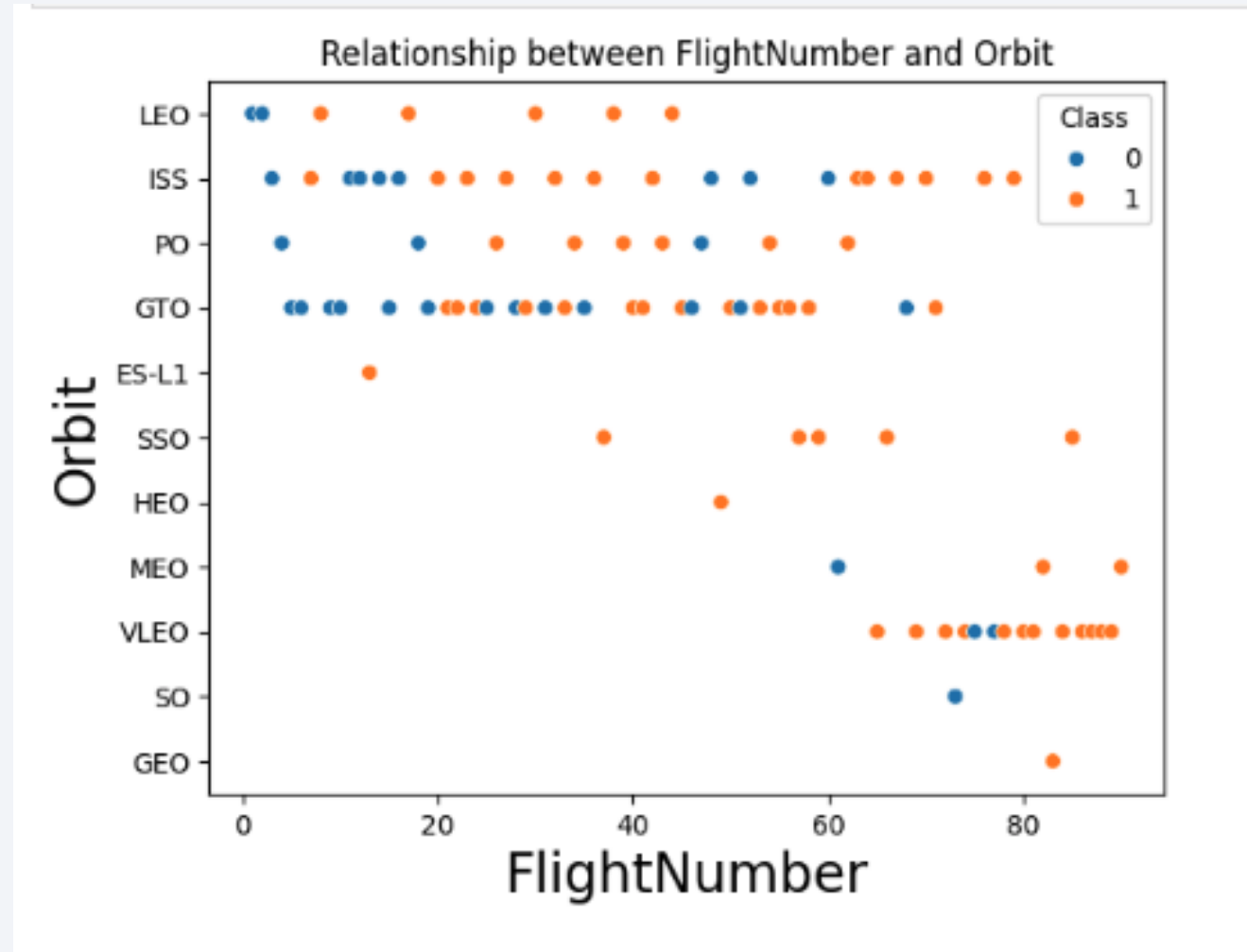
Pay Load Mass Vs. Launch Site

- CCAFS SLC 40 Launch site and KSC LC 39A shows Payload Mass in the entire range (0 -16000) kg
- VAFB SLC 4E site has pay load mass in the range (0 -10000) kg
- All the sites have more missions with payload mass <10000 kg and few missions in the higher range



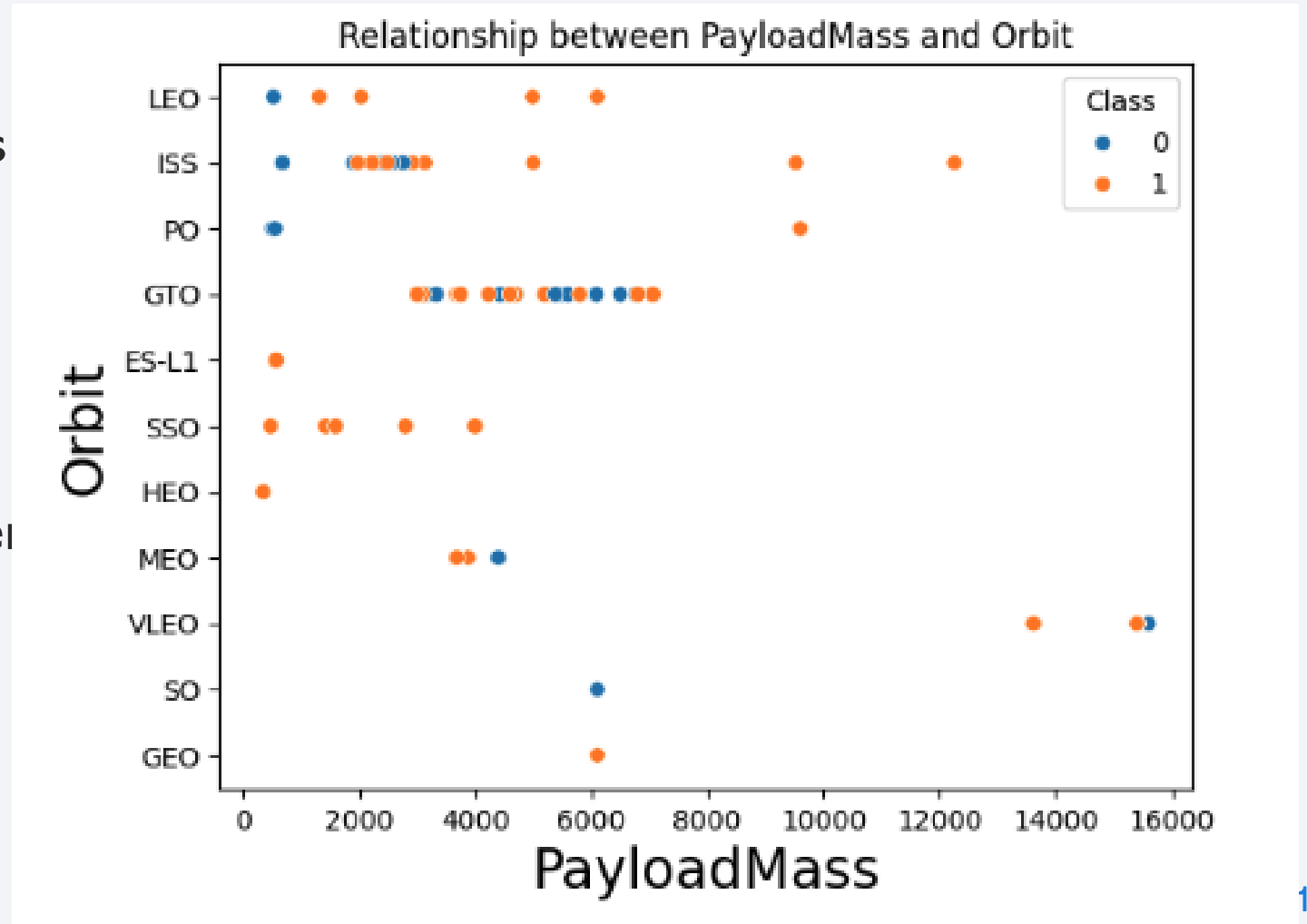
Flight Number vs. Orbit Type

- ISS, SSO, VLEO, LEO have more success rates.
- At lower flight numbers (early missions), there are more failures.
- When the number of flights sent increases (higher flight numbers), the success rate increases.



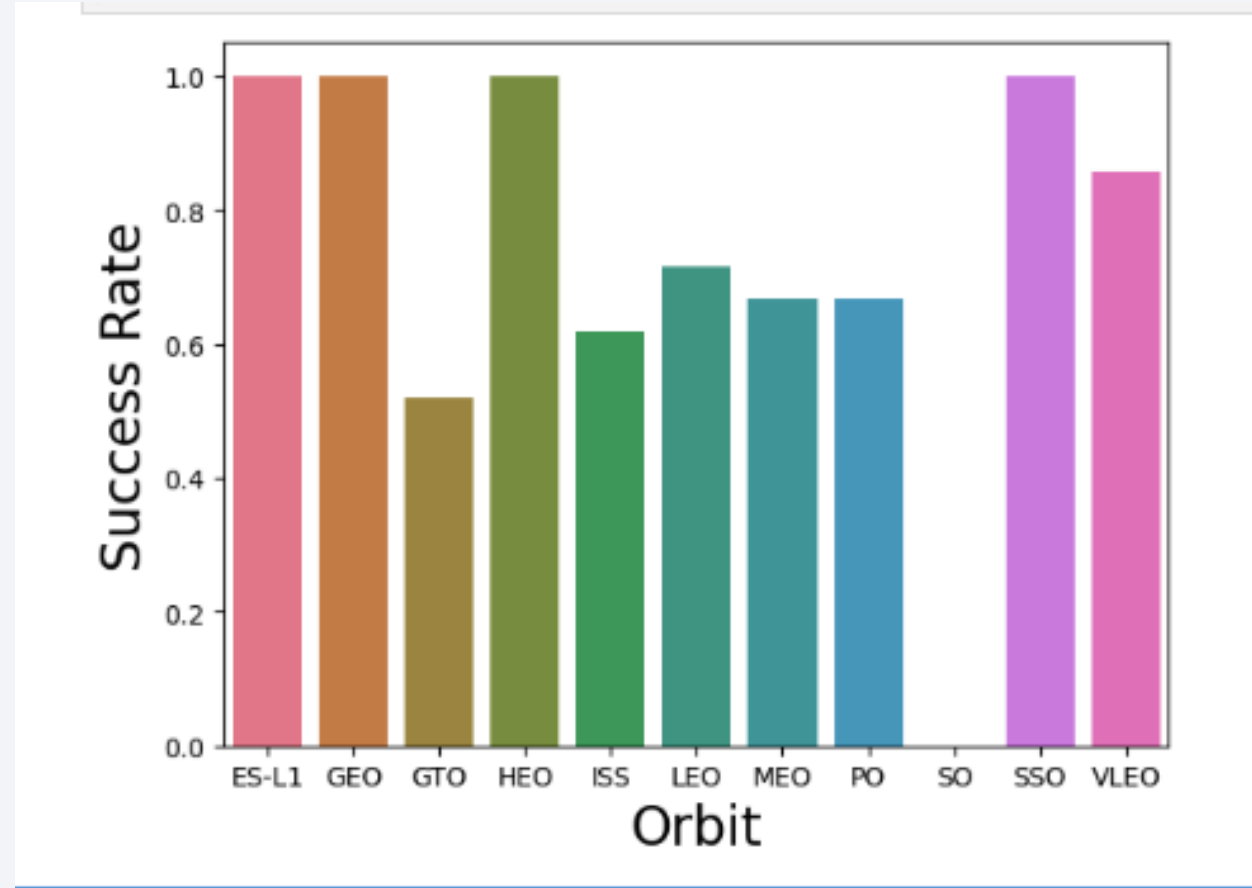
Payload vs. Orbit Type

- LEO has lower payload mass <16000.
- Most of the orbits have payload mass between 0-8000.
- ISS, PO, and VLEO has higher payload mass and their success rate is good.



Success Rate vs. Orbit Type

- GTO orbit has the minimum success rate
- ES-L1, GEO, HEO, SSO orbits have 100% success rates.
- SO orbit has no success



Launch Success Yearly Trend

- The success rate increased with increase in years.
- Years 2017 and 2019 have the higher success rate.
- Between 2013 and 2017, success rate has gradually increased.



A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark blue, with a thin layer of white clouds. A bright, glowing arc of city lights is visible along the horizon, indicating a coastal or urban area. The text "Section 3" is overlaid on the left side of the image.

Section 3

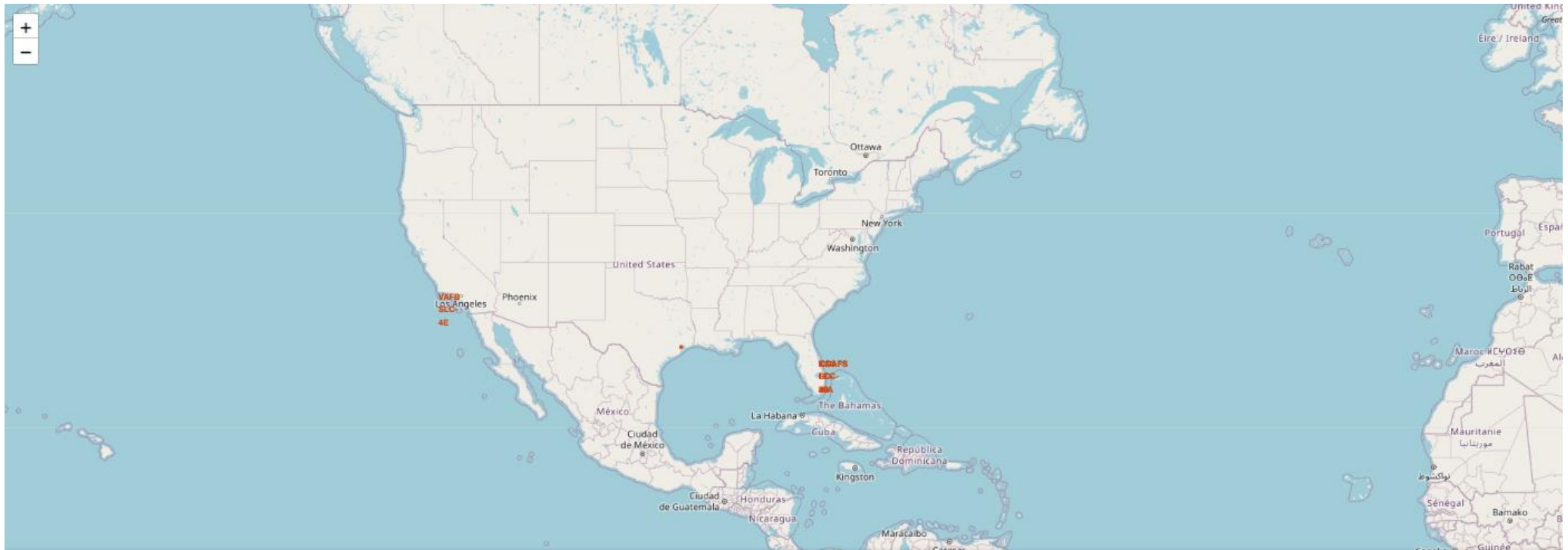
Launch Sites Proximities Analysis

Launch Success Related Location and Proximities

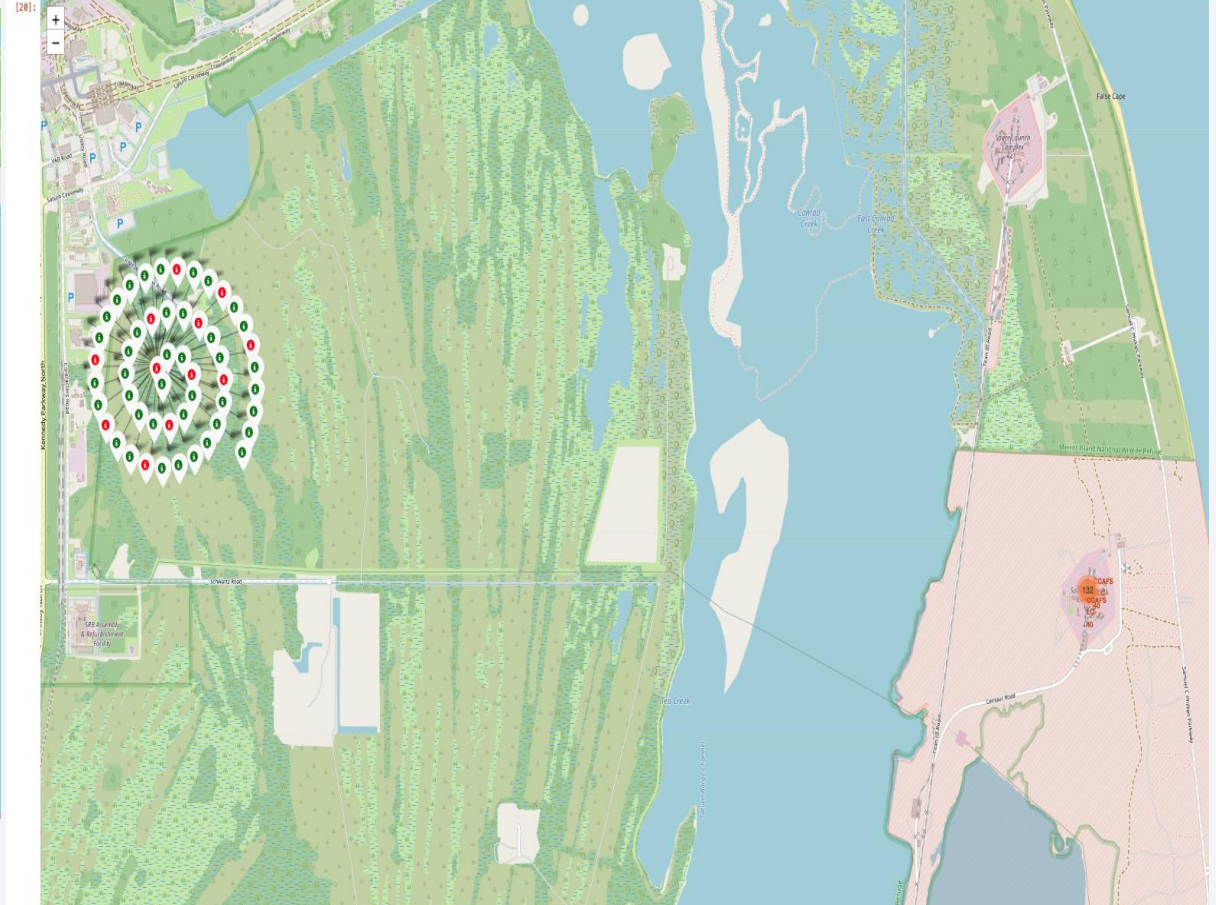
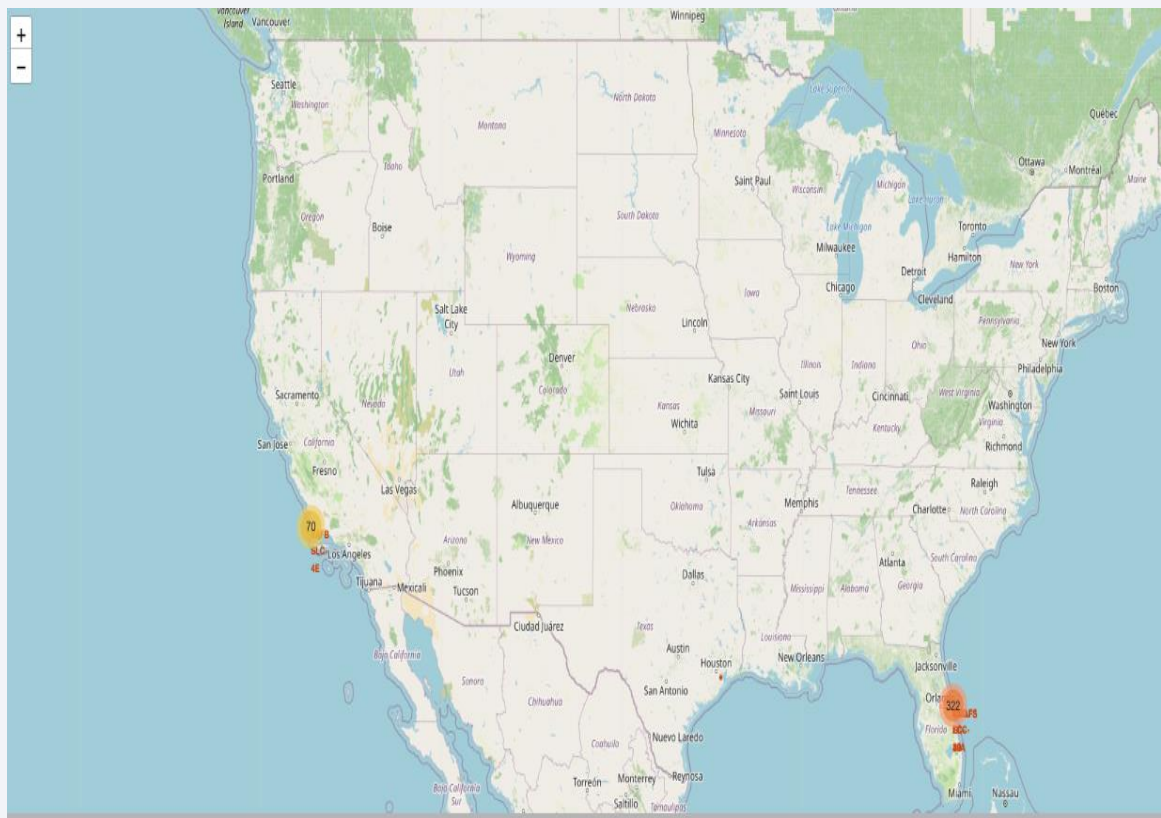
- Interactive visual analytics was performed using Folium
- Relationship between success rate, launch site location, proximities of the launch site to railway line, highways, coastal line, and cities are analyzed.

Launch Sites Locations

- All Launch sites are located on the coastal lines of North America



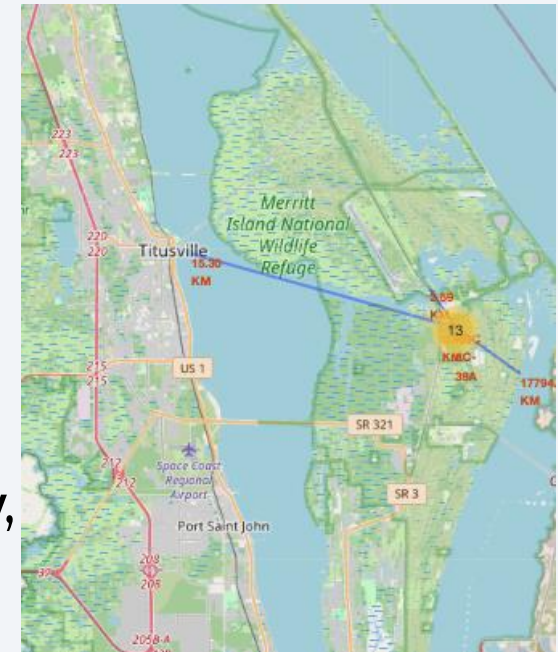
Launch Outcomes



- Success (green), failures (red) at Launch site KSC LC-39A

Build an Interactive Map with Folium

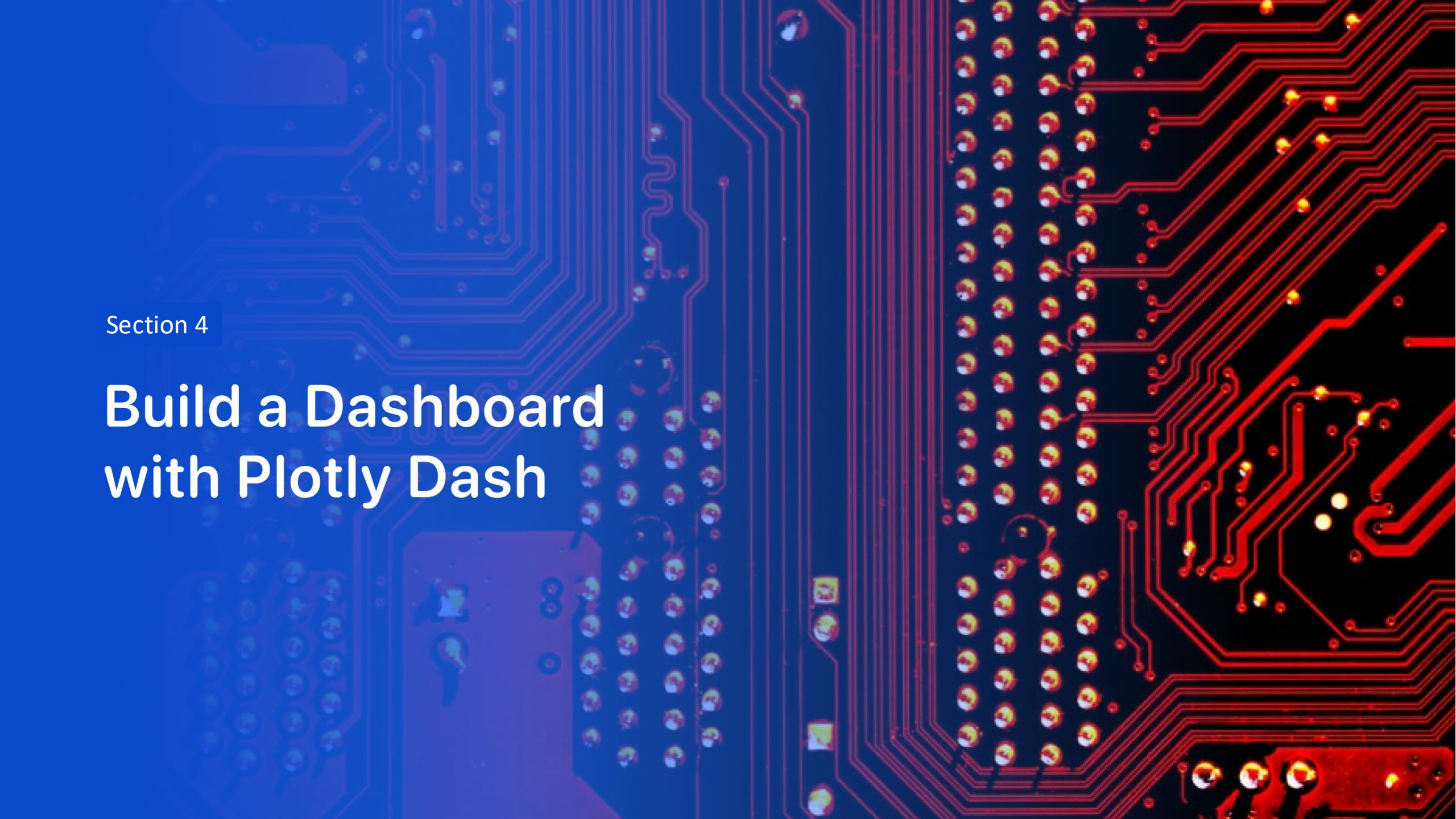
- Folium Map object was created
- Folium circles are added to with a text label on specific coordinate representing each launch site
- Marker color column is created with green color for landing outcome class 1 and red color for class 0.
- Launches from the same site are added as a cluster
- From one of the sites, we find and marked close by railway line, highway, city, and a coastal line
- We added connecting line from the launch site the railway line, highway, city, and coastal line and added on the map.
- We also calculated distance to them and marked on the map
- Reference: https://github.com/gandhiUH/DataScience-Capstone-/blob/main/lab_jupyter_launch_site_location.ipynb



Proximity of KSC LC 39-A

- The proximity of KSC LC 39-A to a railway station, coastline, and highway is shown in the screen shot.





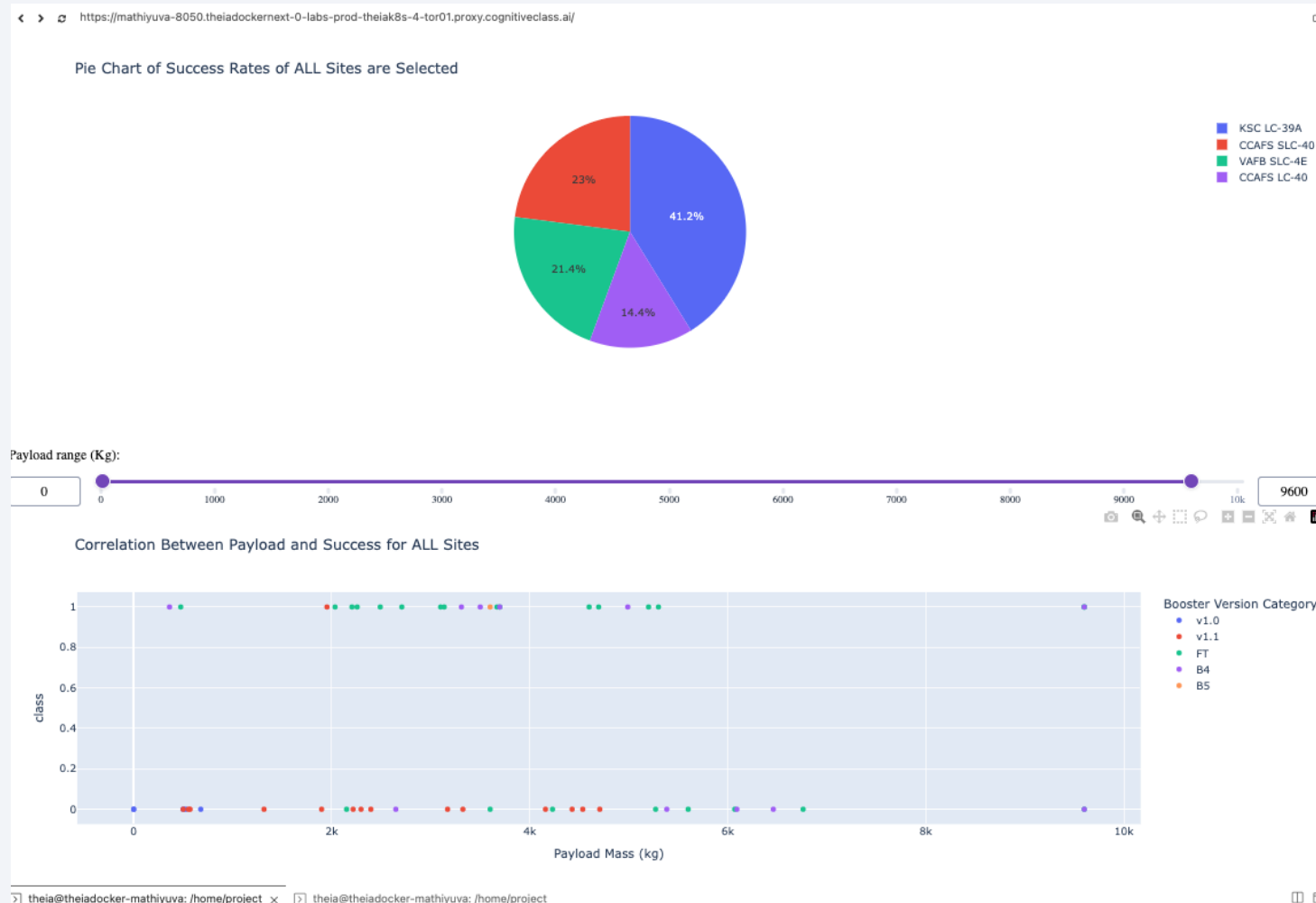
Section 4

Build a Dashboard with Plotly Dash

Build a Dashboard with Plotly Dash

- Plotly Dashboard app is created for interactive visual analytics on SpaceX launch data.
- This dashboard will provide the following insights:
 - The site which has largest successful launches
 - The site that has the highest success rate
 - The payload range that has the lowest launch success rate
 - The payload range that has the highest launch success rate
 - The F9 Booster version that has the highest launch success rate

Build a Dashboard with Plotly Dash



- Top pie chart shows the success rate distribution among the 4 sites
- Bottom scatter plot shows the relationship between the payload mass and success class from all the sites
- FT and V1.0 Booster version shows more success rates
- Most of the success rates are at payload mass less than 6000 kg

Reference: <https://github.com/gandhiUH/DataScience-Capstone-/blob/main/spacex-dash-app-v2.py>

Success Rates of All Launch Sites

- KSC LC-39A Launch site has the highest success rate of 41.2%.
- CCAFS SLC-40 has the success rate of 23%
- VAB SLC-4E's success rate is 21.4%
- CCAFS LC-40 has the lowest success rate of 14.4%

Pie Chart of Success Rates of ALL Sites are Selected



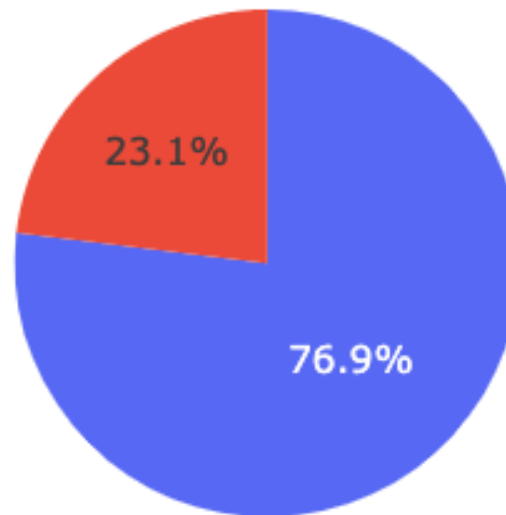
Launch Success Ratio of KSC LC -39A

The Launch Success ratio of the KSC LC-39A site is 76.9% success and 23.1% of failure.

Debug



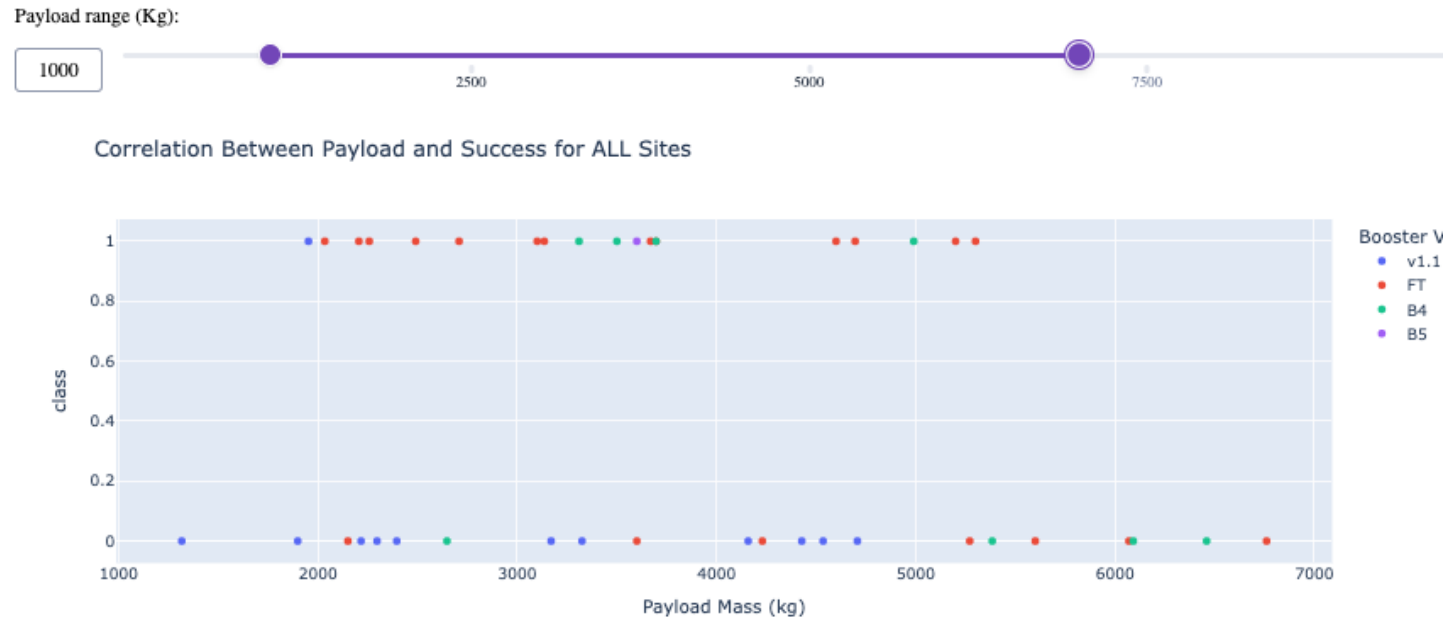
Pie Chart of Success Rates of KSC LC-39A Site is Selected



1
0

Payload Vs. Success Rate

- The top screenshot shows the success rates when payload range is in between 0 -10000 kg.
- The bottom screenshot shows the success rates when payload range is in between 1000 - 7000 kg.
- In both screen shots, the FT Booster version shows the highest success rate



EDA Visualization - Results

- The success rate increased with increase in years.
- Years 2017 and 2019 have the higher success rate.
- GTO orbit has the minimum success rate
- ES-L1, GEO, HEO, SSO orbits have 100% success rates.
- Most of the orbits have payload mass between 0-8000.
- ISS, PO, and VLEO has higher payload mass and their success rate is good.
- ISS, SSO, VLEO, LEO have more success rates.
- At lower flight numbers (early missions), there are more failures.
- When the number of flights sent increases (higher flight numbers), the success rate increases.
- CCAFS SLC 40 Launch site and KSC LC 39A shows Payload Mass in the entire range (0 - 16000) kg

EDA Visualization - Results

- VAFB SLC 4E site has pay load mass in the range (0 -10000) kg
- All the sites have more missions with payload mass <10000 kg and few missions in the higher range
- CCAFS SLC 40 Launch site has more flights sent. The success rate increased as number of flights sent increased.
- VAFB SLC4E site has sent minimum number of flights
- KSC LC-39A Launch site has the highest success rate of 41.2%.
- CCAFS LC-40 has the lowest success rate of 14.4%
- FT and V1.0 Booster version shows more success rates
- Most of the success rates are at payload mass less than 6000 kg
- As the flight number increases, the first stage is more likely to land successfully.
- All Launch sites are located on the coastal lines of North America

Section 5

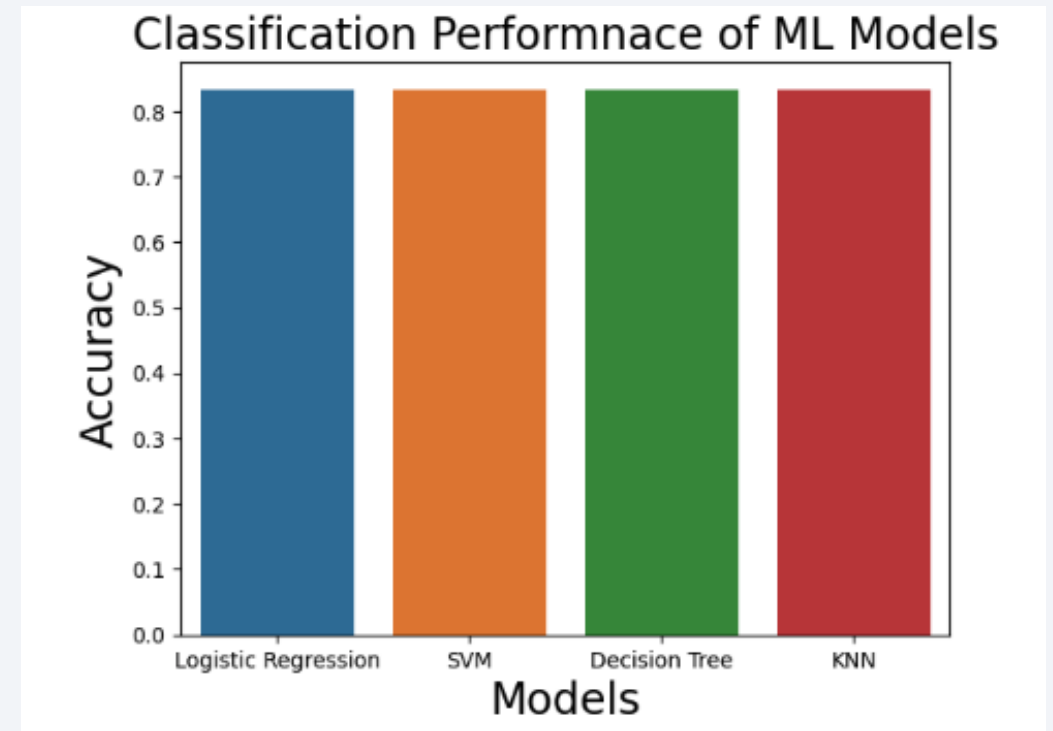
Predictive Analysis (Classification)

Predictive Analysis (Classification)

- The features data is preprocessed with StandardScaler.
- Train and test data sets are split with 0.2 test size and a random state of 2.
- We used Logistic Regression, SVM, Decision Trees, and KNN.
- We used Grid Search CV to fine tune the hyper parameters.
- We found the classification accuracy using the score() method and found to be the same 83.3% from all algorithms.
- We also plotted confusion matrix which is similar from all algorithms.

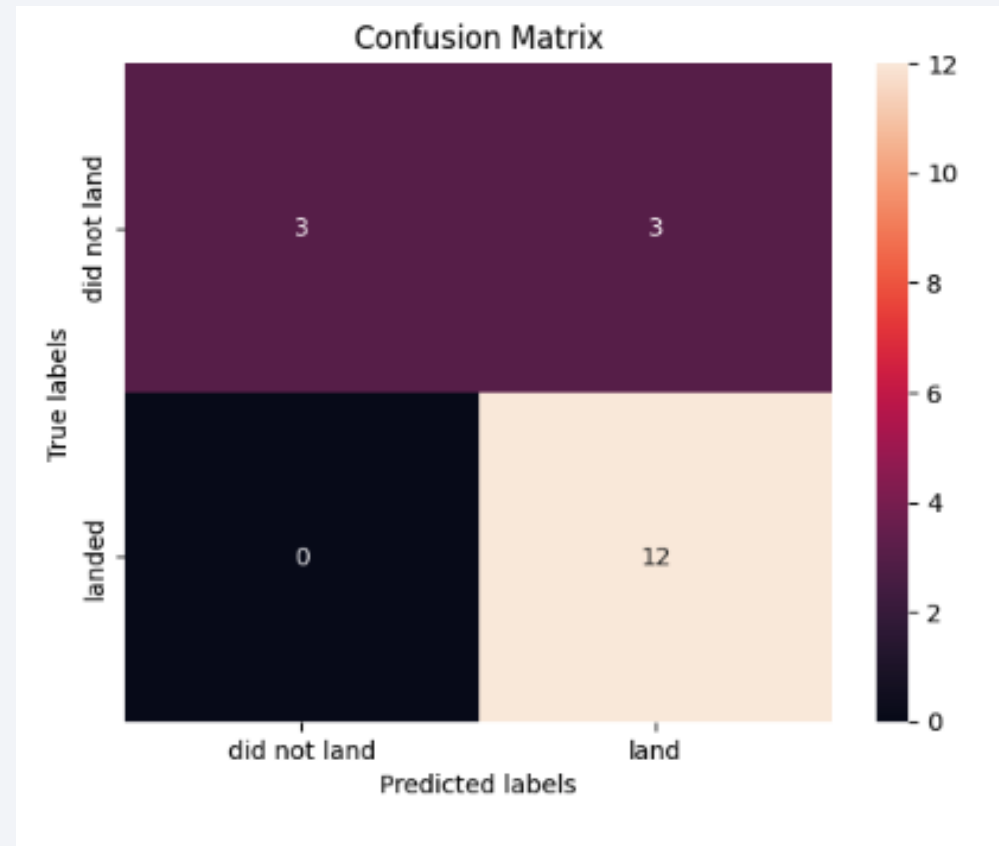
Classification Accuracy

- In this project Logistic Regression, SVM, Decision Tree, and KNN classifiers are used to predict whether the first stage of the rocker launch will land successfully or not.
- The classification accuracy of all the classifier are the same and equal to 83.3%



Confusion Matrix

- Confusion matrix of all the classifiers are the same
- The confusion matrix shows that the model correctly predicted all successful landings and achieved an overall accuracy of 83.3%.
- The model performs very well in predicting successful landings but is less accurate in predicting failed landings.
- Recall is $\frac{TP}{TP+FN} = \frac{12}{12+0} = 1 = 100\%$
- Precision $\frac{TP}{TP+FP} = \frac{12}{12+3} = 0.8 = 80\%$



Conclusions

- SpacX launch data was collected using REST API End points and Python Request's GET method.
- The data is downloaded in json is converted to Pandas DataFrame. Data from tables from the HTML web page is scrapped using BeautifulSoup object.
- Collected data is cleaned and organized (Data Wrangling)
- Exploratory Data Analysis is done using SQL, Dash, and Plotly.
- The data showed there is correlation between success and the number of missions from a site, the more the missions (number of flights), more the success rate
- Success rate has increased between 2013-2019
- KSC LC-39A Launch site has the highest success rate of 41.2% among the given locations with 76.9% success and 23.1% failures.
- Landing success is predicted using Logistic Regression, SVM, Decision Tree, and KNN classifiers.
- All the classifiers achieved the same classification accuracy of 83.3%

Appendix

- Project Github Link:
- <https://github.com/gandhiUH/DataScience-Capstone-.git>

Thank you!

